

Deficit classifications for diagonals of square partial isometries

Major partial answer to Question 6.2 of arXiv:1602.08435

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Status. `candidate_major_partial_likely_valid`.

Abstract

Jasper, Loreaux, and Weiss ask for the diagonals of partial isometries with prescribed kernel and cokernel dimensions, in particular square partial isometries. For a sequence $d = (d_j)$, put

$$\Delta(d) = \sum_j (1 - |d_j|) \in [0, \infty].$$

We prove that every diagonal of a square partial isometry of defect r satisfies $\Delta(d) \geq r$, with the extended interpretation when $r = \infty$. This bound is sharp. It gives a complete classification for infinite defect: precisely $|d_j| \leq 1$ and $\Delta(d) = \infty$. In fact, under infinite deficit every prescribed square defect $r = 0, 1, 2, \dots, \infty$ is realizable. For finite r , we also prove the converse when $\Delta(d) = r$; for positive finite r the converse holds as well whenever only finitely many entries fail to be unimodular. All constructions preserve real scalars for real diagonals.

1 Source question and notation

Let H be a separable infinite-dimensional Hilbert space with fixed orthonormal basis $(e_j)_{j \in \mathbb{N}}$. A partial isometry V is *square of defect r* when

$$\dim \ker V = \dim \ker V^* = r, \quad r \in \{0, 1, 2, \dots, \infty\}.$$

Its diagonal is $d_j = \langle V e_j, e_j \rangle$. Question 6.2 of the source asks for these diagonals; see Figure 1.

We use two results already stated in the source paper [1]: finite-dimensional Thompson's theorem and its unitary-diagonal theorem. The former says that $z_1, \dots, z_N$ are the diagonal of a matrix with singular values $s_1 \geq \dots \geq s_N$ exactly when the Thompson inequalities hold. The latter says that $z = (z_j)$ is the diagonal of a unitary exactly when

$$|z_j| \leq 1, \quad 2(1 - \inf_j |z_j|) \leq \sum_j (1 - |z_j|). \quad (1)$$

2 Idea of the proof

There are two complementary mechanisms. For necessity, look simultaneously at the kernel and cokernel projections. Their diagonal masses cannot both be large at a coordinate where the diagonal entry of the partial isometry is nearly unimodular. Cauchy-Schwarz makes this quantitative and summing recovers the defect.

For sufficiency, total modulus deficit is a budget for zero singular values. On a finite block, one unit of deficit buys one zero singular value through Thompson's theorem. Infinite deficit can

therefore be divided into finite blocks to create infinite defect, or one finite block can create any prescribed finite defect while a unitary realizes the remaining infinite-deficit tail. At exact finite equality, the entire deficit is instead realized as the diagonal of a finite-rank projection, whose complement becomes the initial projection of the desired partial isometry.

3 The sharp necessary inequality

Proposition 1 (Two-sided defect bound). *Let V be a partial isometry with diagonal d . Put*

$$P = I - V^*V, \quad Q = I - VV^*, \quad p_j = \langle Pe_j, e_j \rangle, \quad q_j = \langle Qe_j, e_j \rangle.$$

Then, pointwise,

$$p_j + q_j \leq 2(1 - |d_j|). \quad (2)$$

Consequently, if V is square of finite defect r , then

$$|d_j| \leq 1 \quad \text{and} \quad \Delta(d) \geq r. \quad (3)$$

If both defects are infinite, then $\Delta(d) = \infty$.

Proof. The vectors Ve_j and VV^*e_j lie in the final space of V , and

$$d_j = \langle Ve_j, e_j \rangle = \langle Ve_j, VV^*e_j \rangle.$$

Moreover,

$$\|Ve_j\|^2 = 1 - p_j, \quad \|VV^*e_j\|^2 = \|V^*e_j\|^2 = 1 - q_j.$$

Cauchy–Schwarz followed by the arithmetic–geometric mean inequality gives

$$|d_j| \leq \sqrt{(1 - p_j)(1 - q_j)} \leq 1 - \frac{p_j + q_j}{2},$$

which is (2). If P, Q have rank $r < \infty$, Tonelli’s theorem and the trace formula for positive finite-rank operators yield

$$2r = \sum_j (p_j + q_j) \leq 2 \sum_j (1 - |d_j|).$$

If both ranks are infinite, both projection-diagonal sums diverge, and the same inequality forces $\Delta(d) = \infty$. Contractivity of V gives $|d_j| \leq 1$. $\square$

The use of both defect projections is essential: either one alone gives only the weaker lower bound $r/2$.

4 Finite Thompson blocks

Proposition 2 (Defect block). *Let B be a finite index set of size N , let $1 \leq r \leq N$, and suppose*

$$|d_j| \leq 1 \quad (j \in B), \quad \sum_{j \in B} (1 - |d_j|) \geq r. \quad (4)$$

Then there is an N -dimensional partial isometry of defect r whose diagonal is $(d_j)_{j \in B}$. It can be chosen real when the d_j are real.

Proof. Apply finite Thompson to the singular-value list

$$s = (\underbrace{1, \dots, 1}_{N-r}, \underbrace{0, \dots, 0}_r).$$

Write $b_1 \geq \dots \geq b_N$ for the decreasing rearrangement of the $|d_j|$. For $k \leq N - r$, $\sum_{i \leq k} b_i \leq k = \sum_{i \leq k} s_i$. For $k > N - r$,

$$\sum_{i \leq k} b_i \leq \sum_{i \leq N} b_i = N - \sum_{j \in B} (1 - |d_j|) \leq N - r = \sum_{i \leq k} s_i.$$

The final Thompson inequality follows from

$$\sum_{i=1}^{N-1} b_i - b_N \leq \sum_{i=1}^N b_i \leq N - r = \sum_{i=1}^{N-1} s_i - s_N.$$

Thus Thompson gives a matrix A with diagonal (d_j) and singular values s . Since the spectrum of A^*A is contained in $\{0, 1\}$, A^*A is a projection, so A is a partial isometry. Both defects equal r . Thompson's real conclusion gives the last assertion. $\square$

5 Infinite deficit: a complete classification

Theorem 1 (Defect flexibility). *Suppose $|d_j| \leq 1$ and $\Delta(d) = \infty$. Then, for every*

$$r \in \{0, 1, 2, \dots, \infty\},$$

there is a square partial isometry of defect r with diagonal d . If d is real, it can be realized on a real Hilbert space.

Proof. For $r = 0$, condition (1) is automatic, so the source's unitary theorem applies.

Fix $1 \leq r < \infty$. Choose a finite set B with $\sum_{j \in B} (1 - |d_j|) \geq r$. Proposition 2 gives a defect- r partial isometry on the corresponding finite-dimensional block. Removing finitely many terms leaves infinite deficit, so the complementary sequence satisfies (1) and is the diagonal of a unitary. The orthogonal direct sum has both defects r .

For $r = \infty$, greedily partition $\mathbb{N}$ into finite sets $B_1, B_2, \dots$ with

$$\sum_{j \in B_k} (1 - |d_j|) \geq 1.$$

This is possible because every tail after finitely many removals still has infinite sum. Apply Proposition 2 with $r = 1$ on every block. Their direct sum has infinite-dimensional kernel and cokernel. All ingredients have real versions. $\square$

Combining necessity and sufficiency gives the promised full subcase.

Corollary 1 (Infinite square defect). *A complex sequence d is the diagonal of a square partial isometry with infinite-dimensional kernel and cokernel if and only if*

$$|d_j| \leq 1 \quad \text{for all } j, \quad \sum_j (1 - |d_j|) = \infty.$$

The same characterization holds over the reals for real d .

6 Two exact finite-defect regimes

Theorem 2 (Equality boundary). *Let $r < \infty$. If $|d_j| \leq 1$ and $\Delta(d) = r$, then d is the diagonal of a square partial isometry of defect r .*

Proof. Set $a_j = 1 - |d_j|$. The sequence $a = (a_j)$ lies in $[0, 1]$ and has sum r . By the finite-rank Schur–Horn theorem (equivalently, the finite-trace case of Kadison’s carpenter theorem stated in the source), there is a rank- r projection P with diagonal a . Thus $I - P$ has diagonal $(|d_j|)$. Choose a diagonal unitary $W = \text{diag}(w_j)$ with $w_j|d_j| = d_j$, and put

$$V = W(I - P).$$

Then $V^*V = I - P$ and $VV^* = W(I - P)W^*$; both complementary projections have rank r , and the diagonal of V is d . For real d , the source’s real finite-rank Schur–Horn theorem gives a real P , and $w_j \in \{-1, 1\}$ may be used. $\square$

Theorem 3 (Eventually unimodular sequences). *Fix $1 \leq r < \infty$, and suppose $|d_j| \leq 1$ with $|d_j| = 1$ except at finitely many indices. Then d is the diagonal of a square partial isometry of defect r if and only if $\Delta(d) \geq r$.*

Proof. Necessity is Proposition 1. For sufficiency, let B contain all indices with $|d_j| < 1$. Proposition 2 realizes that finite block with defect r ; the complementary diagonal is itself a diagonal unitary. Their direct sum proves the claim. $\square$

7 Scope, novelty, and remaining obstruction

The complete open question is not claimed. The unresolved regime after these results is finite r with

$$r < \Delta(d) < \infty$$

and infinitely many non-unimodular entries. Finite-section partial isometries need not have a strongly convergent subsequence: off-diagonal unitary mass can escape even when the finite-rank kernel and cokernel projections are tight. Nor can the tail always be delegated to a unitary; superlacunary deficit sequences can violate (1) after every finite removal. These are genuine gaps, not merely omitted details.

Cheap-index searches and bounded arXiv searches through 12 August 2026 found no later solution of Question 6.2 and no statement of Corollary 1. Novelty confidence is moderate: the arguments use classical ingredients and could be unindexed folklore. Human review should focus on Proposition 2, the projection-diagonal use in Theorem 2, and the claim that the selected question remains open in the literature.

References

- [1] J. Jasper, J. Loreaux, and G. Weiss, *Thompson’s theorem for compact operators and diagonals of unitary operators*, arXiv:1602.08435 (2016).

Next consider $U_{1,j}$ with singular values $\sigma_1 \geq \sigma_2$. Note that (5.2) guarantees $d_1 \leq d_j$ for all $j \geq 2$. Then using the singular value decomposition of $U_{1,j}$ and multiplying also by $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, we can multiply U on the left and right by unitaries to get a new unitary whose diagonal is $\mathbf{d}$ with d_1, d_j replaced by $\sigma_1 - \sigma_2$. By (5.2), we have that $\sigma_1 - \sigma_2 \leq d_j - d_1$, and so by Lemma 5.8 $U_{1,j}$ is selfadjoint. Since $U_{j,k}$ is selfadjoint for all $j, k \in \mathbb{N}$, U is selfadjoint. ■

6. OPEN QUESTIONS

Question 6.1. *Characterize the diagonals of normal operators with specified singular values.*

Question 6.2. *Characterize the diagonals of partial isometries with kernel dimension n and co-kernel dimension m ($1 \leq n, m \leq \infty$). In particular, find the diagonals of what we call square partial isometries. An operator A is said to be square if A^*A and AA^* have the same dimension of their kernels.*

Question 6.3. *Characterize the diagonals of positive compact operators with one dimensional kernel, and then those with finite dimensional kernel.*

REFERENCES

[AM07] M. Argerami and P. Massey. “A Schur–Horn theorem in II_1 factors”. In: *Indiana*

Figure 1: Question 6.2 on source page 21 asks for the diagonals of partial isometries with prescribed kernel and cokernel dimensions, especially square partial isometries.